

Fixed point Iteration

Q10

Use algebraic manipulation to show that each of the following functions has a fixed point at p precisely when $f(p) = 0$, where $f(x) = x^4 + 2x^2 - x - 3$.

a. $g_1(x) = (3 + x - 2x^2)^{1/4}$

$\Rightarrow$ Show $g(p) = p$, where $f(p) = 0$

$$g(x) = (3 + x - 2x^2)^{1/4}$$

$$(x)^4 = ((3 + x - 2x^2)^{1/4})^4$$

$$x^4 = \underline{3 + x - 2x^2}$$

$$x^4 + 2x^2 - x - 3 = 0$$

$\Downarrow$

$$f(x) = 0$$

b. $g_2(x) = \left(\frac{x + 3 - x^4}{2}\right)^{1/2}$

$$= g(x) = \left(\frac{x + 3 - x^4}{2}\right)^{1/2}$$

$$(x)^2 = \left(\left(\frac{x + 3 - x^4}{2}\right)^{1/2}\right)^2$$

$$2 \times x^2 = \frac{x + 3 - x^4}{2} \times 2$$

$$2x^2 = \underline{x + 3 - x^4}$$

$$x^4 + 2x^2 - x - 3 = 0$$

$$f(x) = 0$$

$$\checkmark \quad g_3(x) = \left(\frac{x+3}{x^2+2} \right)^{1/2}$$

$$= g(x) = \left(\frac{x+3}{x^2+2} \right)^{1/2}$$

$$(x)^2 = \left(\left(\frac{x+3}{x^2+2} \right)^{1/2} \right)^2$$

$$\frac{x^2}{1} = \frac{x+3}{x^2+2}$$

$$x^2(x^2+2) = x+3$$

$$x^4 + 2x^2 = x+3$$

$$x^4 + 2x^2 - x - 3 = 0$$

$$f(x) = 0$$

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Q28 2. a. Perform four iterations, if possible, on each of the functions g defined in Exercise 1. Let $p_0 = 1$ and $p_{n+1} = g(p_n)$, for $n = 0, 1, 2, 3$.

from Q1 $\rightarrow$ $\checkmark \quad g_1(x) = (3 + x - 2x^2)^{1/4}$

$$p_0 = 1$$

$$p_1 = g(p_0) = (3 + (1) - 2(1)^2)^{1/4} = 1.18920711$$

$$p_2 = g(p_1) = (3 + (1.18920711) - 2(1.18920711)^2)^{1/4} = 1.0820577$$

$$p_3 = g(p_2) = 1.1496714$$

$$p_4 = g(p_3) = 1.10782055$$

Q70

Use a fixed-point iteration method to determine a solution accurate to within 10^{-2} for $x^4 - 3x^2 - 3 = 0$ on $[1, 2]$. Use $p_0 = 1$.

find $g(x)$ and $g'(x)$ &

$$x^4 - 3x^2 - 3 = 0$$

$$(x^4)^{1/4} = (3x^2 + 3)^{1/4}$$

$$x = (3x^2 + 3)^{1/4} \quad \leadsto \quad g(x) = x$$

$$\downarrow$$

$$g(x) = (3x^2 + 3)^{1/4}$$

$$g'(x) = \frac{1}{4} (3x^2 + 3)^{-3/4} \cdot (6x)$$

use Chain Rule $\Rightarrow$

$$\frac{d}{dx} [f(g(x))] = f'(g(x)) \cdot g'(x)$$

$$\frac{d}{dx} [f(g(x))] = f'(g(x)) \cdot g'(x)$$

$$g'(x) = \frac{6x}{4(3x^2 + 3)^{3/4}}$$

find $g'(x)$ w/ interval $[1, 2]$ &

should be less 1

$$g'(1) = \frac{6(1)}{4(3(1)^2 + 3)^{3/4}} = 0.3912711 < 1$$

$$g'(2) = \frac{6(2)}{4(3(2)^2 + 3)^{3/4}} = 0.3935979 < 1$$

find (p) iteration &

$$p_0 = 1$$

$$p_1 = g(p_0) = (3(1)^2 + 3)^{1/4} = 1.5650845$$

$$p_2 = g(p_1) = 1.7935728 > 10^{-2}$$

$$p_3 = g(p_2) = 1.8859437 > 10^{-2}$$

$$p_4 = g(p_3) = 1.9228478 > 10^{-2}$$

$$p_5 = g(p_4) = 1.9375075 \quad |p_5 - p_4| = 0.01, \text{ No it's equal } 10^{-2}$$

$$p_6 = g(p_5) = 1.9433169 \quad |p_6 - p_5| = 0.005 < 10^{-2},$$

Q 88

Use a fixed-point iteration method to determine a solution accurate to within 10^{-2} for $x^3 - x - 1 = 0$ on $[1, 2]$. Use $p_0 = 1$.

① Find $g(x)$ and $g'(x)$:

$$x^3 - x - 1 = 0$$

$$(x^3) = (x+1)^{1/3}$$

$$x = (x+1)^{1/3}$$

$$\downarrow$$

$$g(x) = (x+1)^{1/3}$$

$$g'(x) = \frac{1}{3} (x+1)^{-2/3} \cdot (1)$$

$$\Rightarrow g'(1) = \frac{1}{3(1+1)^{2/3}} = 0,20 > 1$$

$$\Rightarrow g'(2) = \frac{1}{3(2+1)^{2/3}} = 0,16 > 1$$

$$p_0 = 1$$

$$p_1 = g(p_0) = (x+1)^{1/3} = 1.2599210$$

$$p_2 = g(p_1) = 1.3122938$$

$$p_3 = g(p_2) = \underline{1.3223538} \quad |p_3 - p_2| = 0,01 < 10^{-2}$$

$$p_4 = g(p_3) = \underline{1.3246874} \quad |p_4 - p_3| = 0,002 < 10^{-2}$$

2 digit is similar